

Pranav Sai Palla

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SUMMARY

Computer Science graduate with hands-on experience in Python, machine learning, and large-scale data analysis. Built end-to-end ML systems for classification and prediction — data cleaning, feature engineering, model training, and evaluation — on large public datasets, with results published in IEEE Xplore. Comfortable on Linux, with experience in data pipelines, containerized workflows, and turning telemetry and raw data into actionable, well-documented insights. Strong analytical skills, attention to detail, and a track record of debugging and problem-solving.

EDUCATION

Pittsburg State University — Pittsburg, KS Aug 2025 – May 2026
Bachelor of Technology in Computer Science • GPA: 3.71 / 4.0 • Transfer student (3+1 program)

Malla Reddy University — Hyderabad, India Sept 2022 – May 2025
Bachelor of Technology in Computer Science • GPA: 8.54 / 10.0 • 3+1 program (home institution)

TECHNICAL SKILLS

Programming: Python (primary), SQL, Bash, Java, Go, JavaScript

Machine Learning: Classification & Prediction, XGBoost, Feature Engineering, SMOTE, Model Evaluation (ROC-AUC), NLP, RAG, Vector Search

Data & Analysis: pandas, NumPy, scikit-learn, large-dataset analysis, trend analysis, statistical methods

Tools & Frameworks: FastAPI, LangChain, Streamlit, Node.js/Express, REST APIs

Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, CI/CD, Git/GitHub

Databases: PostgreSQL (pgvector), MongoDB, MySQL, Supabase

Operating Systems: Linux / Unix

PROJECTS

SmartDiab — ML Classification & Prediction Platform

github.com/pranav2910/Smart_Diab

- Developed and implemented a machine learning algorithm for rapid classification and prediction, integrated into an end-to-end software toolset (React, FastAPI, Express, MongoDB) for usability.
- Engineered a full data pipeline — cleaning, SMOTE balancing, feature normalization and selection — and analyzed the large-scale NHANES dataset to surface trends linked to risk.
- Achieved ROC-AUC 0.91 and 87.7% accuracy with XGBoost; authored an IEEE-published paper documenting methodology and validating model performance.

Enterprise AI Handbook Generator — Data Pipeline & RAG System

github.com/pranav2910/handbook-generator

- Built a Python data-ingestion pipeline with LangChain and pdfplumber, parsing unstructured PDFs and optimizing chunking for high-fidelity semantic retrieval via pgvector/Supabase.
- Designed a modular architecture to transform raw documents into 20,000+ word structured outputs, with metadata tracking and systematic documentation of the workflow.
- Iteratively tested a Streamlit interface for upload, generation, and download — validating data quality and user-facing behavior.

Learning Management System — Backend Platform

github.com/pranav2910/CSB1_Learning_Management_System_Backend

- Built backend services with Java Spring Boot and RESTful APIs, applying modular code organization, normalized schemas, and automated end-to-end workflow testing.

EXPERIENCE

Impact Lab YLP Intern — GGI, Hyderabad, India Oct 2024 – Apr 2025

- Analyzed business and governance case studies using AI-driven platforms, researching data, identifying trends, and developing data-backed frameworks to resolve operational bottlenecks.
- Collaborated in cross-functional cohorts, practicing systematic problem decomposition and clear documentation of findings to support data-based decisions.

PUBLICATIONS

- SmartDiabX: An End-to-End AI System for Predicting Undiagnosed Type II Diabetes* — Published in IEEE Xplore, 2025.
ieeexplore.ieee.org/document/11465615